

Calculating Consumer Surplus

- advantage of using a structural model

inclusive value == expected utility of a consumer prior to observing the logit shocks ε_{ij} but knowing that choice will be to maximize utility

Given ε_{ij} is i.i.d. extreme value,

the inclusive value is =

$$\ln \left(\sum_{j \in A} \underbrace{\exp \{x_j \beta_i - \alpha_i p_j + \xi_j\}}_{\equiv V_{ij}} \right)$$

$$CV_i = \frac{\ln \left[\sum_{j=0}^J V_{ij}^{post} \right] - \ln \left[\sum_{j=0}^J V_{ij}^{pre} \right]}{\alpha_i^*},$$

<Recall: *Compensating variation* is income necessary to compensate consumers after the price change. *Equivalent variation* uses the original prices as the base>

Then the mean change in consumer surplus is:

$$M \int CV_i dP_D^*(D) dP_v^*(v),$$